

Fuels

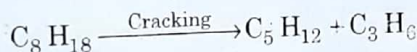
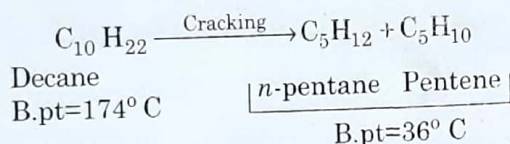
with the help of cooling coils. For example when crude oil is made to pass through first column, the temperature of which is so maintained that fuel oil condense at the bottom and the vapour of other components are passed on to second fractionating column. In the second column, the temperature is so maintained that vapours of gas oil condense at the bottom and so on. Steam is injected at the bottom of each fractionating column for its partial pressure effect. Vapour of gasoline finally pass out from the top (Fig. 1.13).

In top flashing, there is better control of product composition, but requires more pumps and instruments and hence is an expensive process.

1.5.5 Cracking

Of all the fractions obtained by fractional distillation of crude petroleum, gasoline is considered to be the most important fraction. With the rapid growth of automobile industry, the consumption of petrol has increased many folds, but the yield of this fraction is only 20% of the crude oil. Also the quality of the gasoline obtained is not good. This gasoline is called straight run gasoline. On the other hand, the yield of heavier petroleum fractions is quite high. Thus, it is necessary to make better use of these heavy fractions, by converting them into more useful fraction, gasoline. This is achieved by a technique called 'cracking'.

'Cracking' is the process by which heavier fractions are converted into lighter fractions by the application of heat, with or without catalyst. Cracking involves the rupture of C—C and C—H bonds in the chains of high molecular weight hydrocarbons. For example—



Nearly 50% of today's gasoline is obtained by cracking. It may be emphasized here, that the gasoline obtained by cracking is far more superior to the straight run gasoline. The process of cracking involves the following chemical changes:

- Higher hydrocarbons are converted to lower hydrocarbons by C—C cleavage. The products obtained on cracking have low boiling points than initial reactant.
- Formation of branched chain hydrocarbons takes place from straight chain alkanes.
- Unsaturated hydrocarbons are formed from saturated hydrocarbons.
- Cyclization may take place.

Though, the chief application of commercial cracking is the production of gasolines but it is also used for the production of olefins from naphthas, oil gas from kerosene and the reduction of viscosity of straight run residues. Cracking can be carried out by two methods:

- Thermal Cracking:** When cracking takes place simply by the application of heat and pressure, the process is called thermal cracking. The heavy oils are subjected to high temperature and pressure, when the bigger hydrocarbons break down to give smaller molecules of paraffins, olefins etc. The thermal stability among the constituents of petroleum fractions increases as Paraffins < naphthenes (alicyclics) < aromatics. The products obtained, depend upon the operating conditions. Thermal cracking may be carried out either in liquid phase or vapour phase.

- Liquid phase thermal cracking:** By this method, any type of oil (residues, heavy oil) can be cracked. The charge is kept in the liquid form by applying high pressures of

poor quality
straight run gasoline
↑
Lower yield of gasoline
Higher yield of other heavy petroleum products.

Fractional distillation results in

By applying heat and pressure without catalyst.
Liq.
Vapour.

the range 30 kg/cm^2 to 100 kg/cm^2 at a suitable temperature of $476-530^\circ \text{C}$. The cracked products are separated in a fractionating column. The important fractions are: Cracked gasoline (30-35%); Cracking gases (10-45%); Cracked fuel oil (50-55%).

- The cracked gasoline has an octane number of about 70. The cracking gases constitute raw materials for petrochemicals. The cracking residue is used as a fuel.
- (b) Vapour phase thermal cracking: By this method, only those oils which can vaporize at low temperatures can be cracked. The petroleum fractions of low boiling range like kerosene oil, are heated at a temperature of $670-720^\circ \text{C}$ under low pressure of $10-20 \text{ kg/cm}^2$. The time required for cracking in this process is much less as compared to liquid phase thermal cracking. Petrol obtained from vapour phase cracking has better antiknock properties but poorer stability.

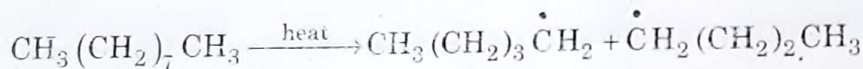
In general, thermal cracking produces petrols which are less stable than straight run petrols due to the presence of highly reactive unsaturated hydrocarbons in them.

Table 1.6 Comparison of liquid and vapour phase thermal cracking.

Characteristic	Liquid Phase-Thermal Cracking	Vapour Phase-Thermal Cracking
Cracking temperature	$475 - 530^\circ \text{C}$	$670 - 720^\circ \text{C}$
Pressure	100 kg/cm^2	$10 - 20 \text{ kg/cm}^2$
Octane rating of Petrol	65 - 70	> 70 ✓
Type of oil	Any heavy oil can be cracked	Oil should be readily vapourized

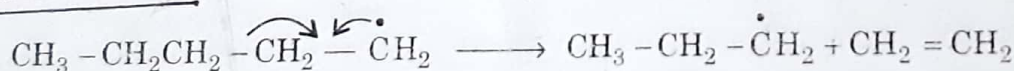
Mechanism of thermal cracking: Thermal cracking is believed to follow free radical mechanism. The mechanism can be illustrated by taking the example of n-nonane.

Initiation

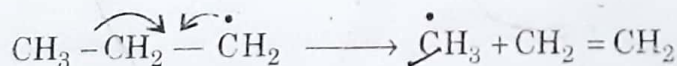


Propagation

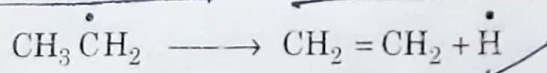
The free radicals formed are thermally unstable and undergo fission at the β -position to yield a new radical and an olefin.



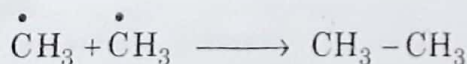
This β scission continues, till a thermally more stable radical is formed.



The radicals having no C—C bond in the β position give rise to hydrogen free radicals.



Termination



2 **Catalytic Cracking:** In this method the cracking is brought about in the presence of a catalyst at much lower temperatures and pressures. The catalyst used is usually a mixture of silica and alumina (hydrated aluminium silicate). The most recent catalyst used is zeolite. The quality and yield of gasoline produced by catalytic cracking is greatly improved. The catalytic cracking has following advantages over the thermal cracking :

- (i) High temperatures and pressures are not required in presence of a catalyst and hence there is saving in the production costs.
- (ii) The use of catalyst not only accelerates the cracking reactions (decomposition, dehydrogenation, isomerization and polymerization), but also introduces new reactions which considerably modify the yield and the nature of the products.
- (iii) The yield of the gasoline is higher. About 70% of raw material is converted into gasoline.
- (iv) The gasoline obtained by catalytic cracking contains low percentage of olefins and high percentage of isoparaffin and aromatic hydrocarbons. This increases the antiknock properties of gasoline. Gasoline formed by catalytic cracking has an octane number of 80.
- (v) No external fuel is required for cracking, since the heat required is obtained by burning off the coke deposited on the catalyst itself, during regeneration process.
- (vi) The process can be better controlled than that of thermal process, so desired products can be obtained.
- (vii) The product contains a very little amount of undesirable sulphur, because a major portion of it escapes out as H_2S gas, during cracking.
- (viii) The gasoline formed contains much less gum and gum forming compounds.
- (ix) Compared with the thermal process, the catalytic cracking yields less coke, less gas and more liquid products.
- (x) The evolution of by-product gas can be further minimized, thereby increasing the yield of desired petrol.
- (xi) Catalysts are selective in action and hence cracking of only high boiling fractions takes place.
- (xii) Coke forming materials are absorbed by the catalysts as soon as they are formed.

Depending upon how the catalyst is used, the catalyst cracking is of two types:

(i) **Fixed bed catalytic Cracking:** In this type of cracking the catalyst is in the form of granules or pellets. These pellets or granules are used in the form of fixed beds in catalyst towers.

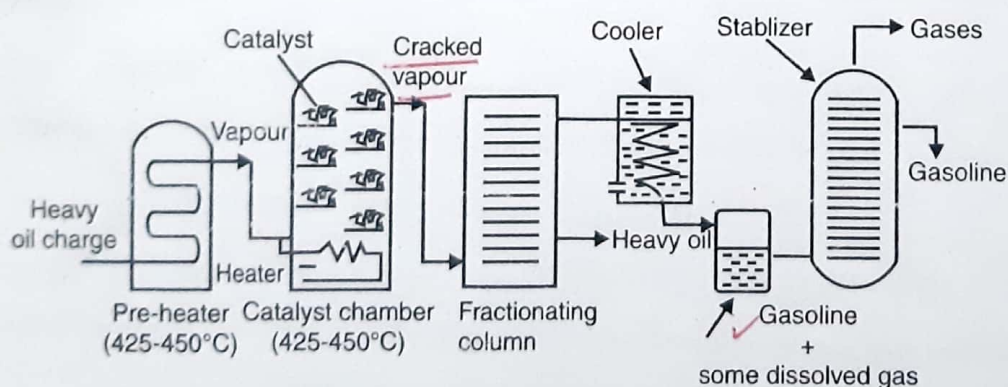


Fig. 1.14 Fixed-bed type catalytic cracking

the range 30 kg/cm² to 100 kg/cm² at a suitable temperature of 476–530° C. The cracked products are separated in a fractionating column. The important fractions are: Cracked gasoline (30-35%); Cracking gases (10-45%); Cracked fuel oil (50-55%).

The cracked gasoline has an octane number of about 70. The cracking gases constitute raw materials for petrochemicals. The cracking residue is used as a fuel.

- (b) Vapour phase thermal cracking: By this method, only those oils which can vaporize at low temperatures can be cracked. The petroleum fractions of low boiling range like kerosene oil, are heated at a temperature of 670-720° C under low pressure of 10-20 kg/cm². The time required for cracking in this process is much less as compared to liquid phase thermal cracking. Petrol obtained from vapour phase cracking has better antiknock properties but poorer stability.

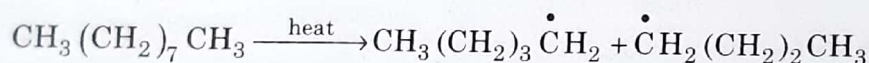
In general, thermal cracking produces petrols which are less stable than straight run petrols due to the presence of highly reactive unsaturated hydrocarbons in them.

Table 1.6 Comparison of liquid and vapour phase thermal cracking.

Characteristic	Liquid Phase-Thermal Cracking	Vapour Phase-Thermal Cracking
Cracking temperature	475 – 530°C	670 – 720°C
Pressure	100 kg/cm ²	10 – 20 kg/cm ²
Octane rating of Petrol	65 – 70	> 70
Type of oil	Any heavy oil can be cracked	Oil should be readily vapourized

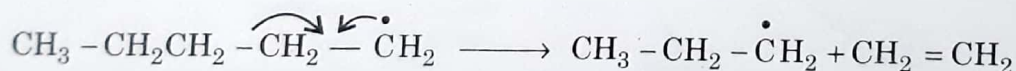
Mechanism of thermal cracking: Thermal cracking is believed to follow free radical mechanism. The mechanism can be illustrated by taking the example of n-nonane.

Initiation



Propagation

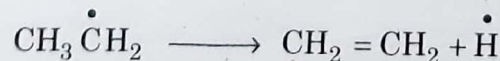
The free radicals formed are thermally unstable and undergo fission at the β-position to yield a new radical and an olefin.



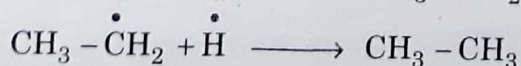
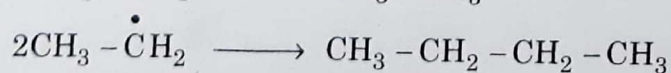
This β scission continues, till a thermally more stable radical is formed.



The radicals having no C—C bond in the β position give rise to hydrogen free radicals.



Termination



2. **Catalytic Cracking:** In this method the cracking is brought about in the presence of a catalyst at much lower temperatures and pressures. The catalyst used is usually a mixture of silica and alumina (hydrated aluminium silicate). The most recent catalyst used is zeolite. The quality and yield of gasoline produced by catalytic cracking is greatly improved. **The catalytic cracking has following advantages over the thermal cracking :**

- (i) High temperatures and pressures are not required in presence of a catalyst and hence there is saving in the production costs.
- (ii) The use of catalyst not only accelerates the cracking reactions (decomposition, dehydrogenation, isomerization and polymerization), but also introduces new reactions which considerably modify the yield and the nature of the products.
- (iii) The yield of the gasoline is higher. About 70% of raw material is converted into gasoline.
- (iv) The gasoline obtained by catalytic cracking contains low percentage of olefins and high percentage of isoparaffin and aromatic hydrocarbons. This increases the antiknock properties of gasoline. Gasoline formed by catalytic cracking has an octane number of 80.
- (v) No external fuel is required for cracking, since the heat required is obtained by burning off the coke deposited on the catalyst itself, during regeneration process.
- (vi) The process can be better controlled than that of thermal process, so desired products can be obtained.
- (vii) The product contains a very little amount of undesirable sulphur, because a major portion of it escapes out as H_2S gas, during cracking.
- (viii) The gasoline formed contains much less gum and gum forming compounds.
- (ix) Compared with the thermal process, the catalytic cracking yields less coke, less gas and more liquid products.
- (x) The evolution of by-product gas can be further minimized, thereby increasing the yield of desired petrol.
- (xi) Catalysts are selective in action and hence cracking of only high boiling fractions takes place.
- (xii) Coke forming materials are absorbed by the catalysts as soon as they are formed.

Depending upon how the catalyst is used, the catalytic cracking is of two types:

- (i) **Fixed bed catalytic Cracking:** In this type of cracking the catalyst is in the form of granules or pellets. These pellets or granules are used in the form of fixed beds in catalyst towers.

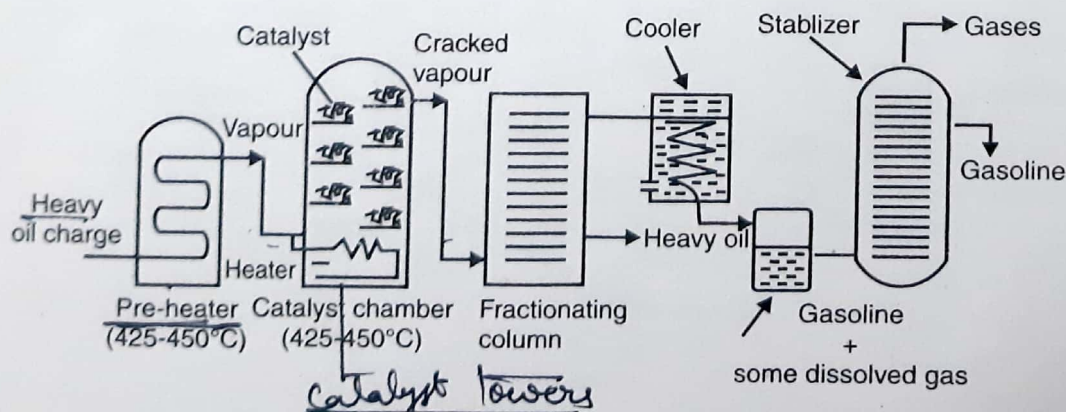


Fig. 1.14 Fixed-bed type catalytic cracking

The heavy oil charge is passed through a heater and heated to a cracking temperature (420-450°C). The charge is then passed through a catalyst chamber. The catalyst used is either silica-alumina gel ($\text{SiO}_2 \cdot \text{Al}_2\text{O}_3$) or bauxite, mixed with clay and zirconium oxide. The catalyst tower is also maintained at 420-450°C and at a pressure of 1 to 5 kg/cm². Cracking of the heavy oil vapours takes place in the catalytic chamber and about 30-40% of the charge is converted into low molecular weight hydrocarbons. The carbon formed (about 4%) during the cracking process gets adsorbed on the catalyst thereby deactivating it. The cracked vapours are passed through the fractionating column, where the gasoline vapours are recovered from the top and the heavy oil gets condensed at the bottom. The gasoline vapours are subjected to a cooler where gasoline along with some other gaseous products gets condensed. Thereafter the gasoline is sent to a stabilizer, where the dissolved gases are removed from the top and pure gasoline is obtained, having the desired boiling range (Fig. 1.14).

During the process, the catalyst stops working after about 10-11 hours, due to the deposition of carbon particles formed during the cracking process. This is reactivated by burning off the carbon in a stream of hot air. During the reactivation of the catalyst, the vapours of heavy oil are diverted through another stand by catalytic chamber so that the reaction proceeds continuously.

As the cracking reactions are endothermic, the temperature in the catalytic chamber is maintained by circulating molten salt (mixture of sodium nitrate and sodium nitrite) in pipes passing through the catalyst bed.

- (ii) **Moving bed Catalytic Cracking:** This is also known as fluid bed catalytic cracking. In this type of cracking, the catalyst is in the form of fine powder which flows down to the cracking chamber, through a hopper. The catalyst gets mixed with the heavy oil vapours and behave much like a fluid which can be circulated in gas streams. The plant for fluid bed or moving bed catalyst cracking consists of a reactor and a regenerator placed side by side (Fig. 1.15)

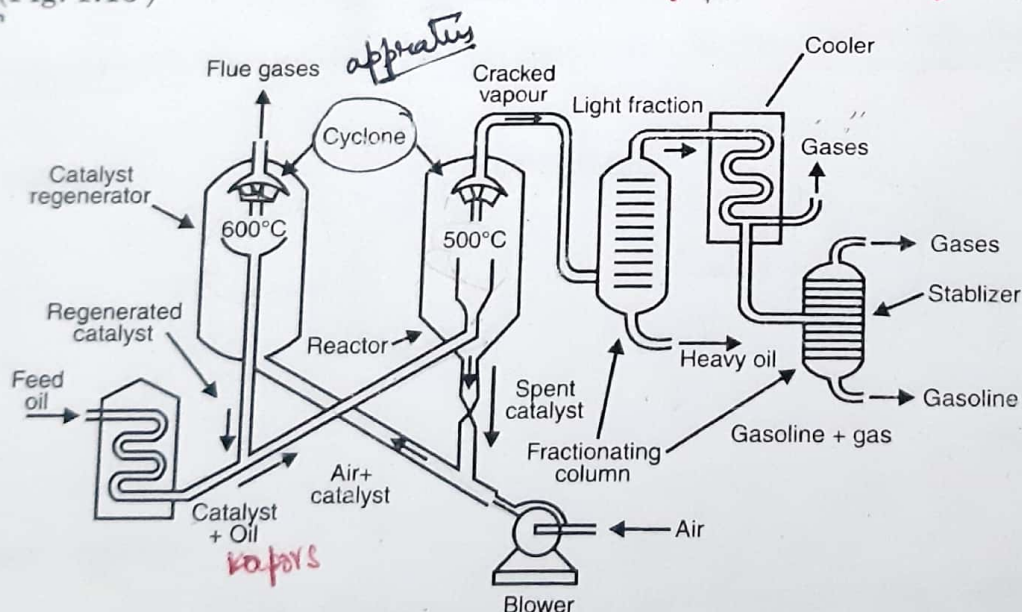
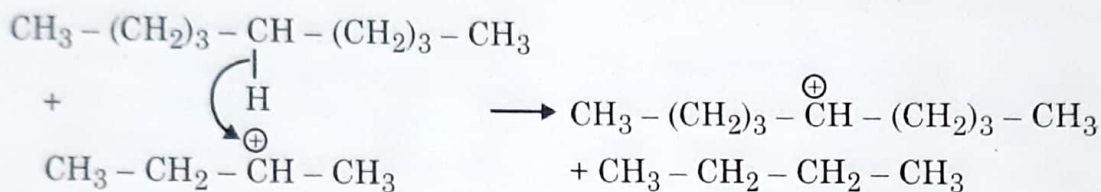
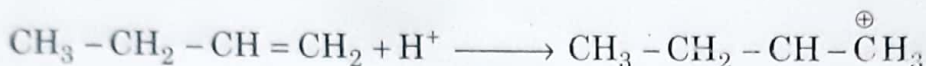
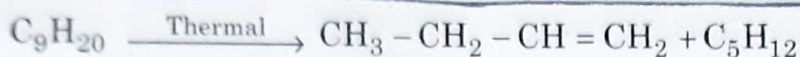


Fig. 1.15 Moving-bed type catalytic cracking

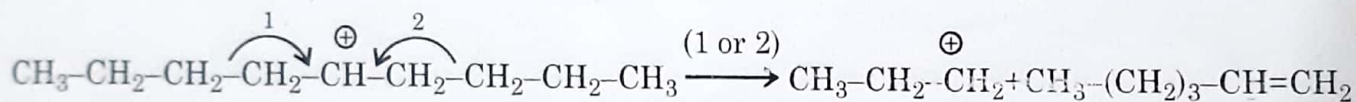
The heavy oil is heated in a heater and the vapours are allowed to mix with the hot catalyst, in the form of fine powder, which is introduced from the regenerator. The mixture is forced into a large reactor maintained at 500°C. Due to the high temperature of the catalyst (600°C), the cracking of the heavy oil starts even before its entry into the reactor. This facilitates the cracking

of the high boiling fractions also. After cracking of the heavy oil in the reactor, the cracked vapours are allowed to pass on to the fractionating column, through the internal cyclone. The internal cyclone is fixed at the top of the reactor which allows only the cracked oil vapours to pass on to fractionating column, retaining the catalyst powder in the reactor itself through the centrifugal action. The catalyst powder gradually becomes heavier due to the deposition of carbon and settles at the bottom. The spent catalyst is forced by an air blast to the regenerator maintained at 600°C . In regenerator carbon is burnt off and catalyst is reactivated. The reactivated catalyst then flows down through a stand pipe to get mixed with a fresh charge. The cyclone fixed at the top of the regenerator allows only the gases such as CO_2 etc to pass, retaining the catalyst particles. The heat liberated during the regeneration process is used to preheat the catalyst.

Mechanism: Catalytic cracking is believed to involve carbonium-ion intermediates. This can be illustrated by taking the example of n-nonane. First, due to thermal cracking, small amount of alkenes is produced. The alkene produced is thus protonated by the catalyst.



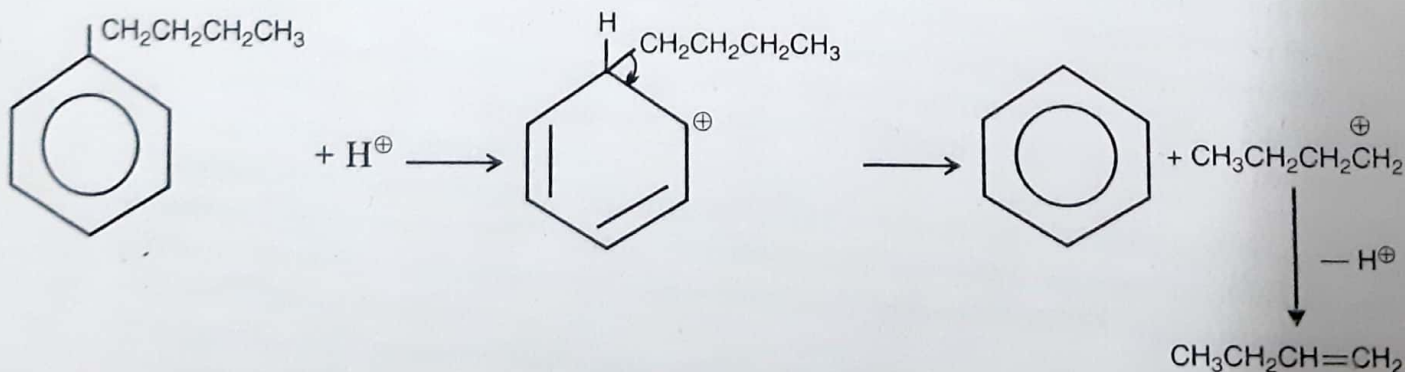
The nonyl carbonium ion then undergoes β cleavage to yield an alkene and a smaller carbonium ion.



The primary carbonium ion rearrange to the more stable secondary carbonium ion by 1,2 hydride shift.



In aromatic hydrocarbons, the rings remain intact and only the side chains are lost.



Knocking is The sharp ⁽³⁾ metallic sound similar to rattling of hammer which is produced in the internal combustion engine due to immature ignition of air/fuel mixture.

FUELS 27

Factors on which knocking depends :

- (i) Engine design
- (ii) Running conditions and
- (iii) Chemical structure of the fuel hydrocarbons. For instance :
 - knocking tendency decreases in the order —

n -alkanes > mono-substituted alkanes > cyclo alkanes > alkenes
> poly substituted alkenes > aromatics.

And for straight chain hydrocarbons, the tendency to knock increases with molecular weight and boiling point, e.g.,

n -hexane > n -pentane > n -butane.

Aromatic hydrocarbons have higher antiknocking properties than paraffins and olefins.

9 OCTANE NUMBER

Knocking Property of the fuel is expressed in Octane numbers. [GGSIPU, IInd Term 2000, 02, 05 ; End Term 2000, 02, 06, 12]

The resistance offered by gasoline to knocking cannot be defined in absolute terms. It is generally expressed on an arbitrary scale, known as **Octane rating** (proposed by Graham Edgar in 1926).

Fuel	Octane number	Characteristics
<i>(More knocking)</i> n -heptane	0	Knocks severely
<i>(Less knocking)</i> iso-octane	100	High resistance to knocking

The % of iso-octane in the n -heptane — iso-octane blend which has the same knocking characteristics as the gasoline sample, under the same set of conditions is known as **octane number**.

9.1 Influence of Structure of the Fuel on the Octane Rating

[GGSIPU, IInd Term 2002 ; End Term 2000, 2006]

(i) As straight chain (or normal) hydrocarbons knock severely in petrol engines so they have lowest octane numbers.

(ii) As the knocking tendency of n -alkanes increases with the increase in the length of carbon chain so anti-knocking properties (and thus octane numbers) also decrease with increase in the length of the carbon chain.

For example :

n -paraffin	Octane number
Methane	122
Ethane	101
Propane	96
n -butane	89
n -pentane	60
n -hexane	29
n -heptane	0
n -nonane	-45

In the same engine under the same set of conditions and the same critical C_r, various blends of n -heptane and iso-octane are burnt and % of iso-octane by volume in blend that knocks under these conditions is octane number of gasoline.

(iii) Branched chain paraffins have lower tendency to knock so they have higher anti-knock properties (and thus octane numbers) in comparison to normal chain paraffins.

Furthermore, resistance to knock increases with the number of branches and their position.

For example :

Hydrocarbon	n-heptane	2-methyl hexane	2, 2-dimethyl pentane
Octane number	0	55	80 ✓

Recently, the hydrocarbon 2,2,3-trimethylpentane called triptane has been assigned an octane number, 125.

(iv) Knocking tendency decreases with the increase in the number of carbon-carbon double bonds in the hydrocarbon fuel.

Thus, olefins have higher anti-knock properties (and thus octane number) than the paraffins of same carbon-chain length.

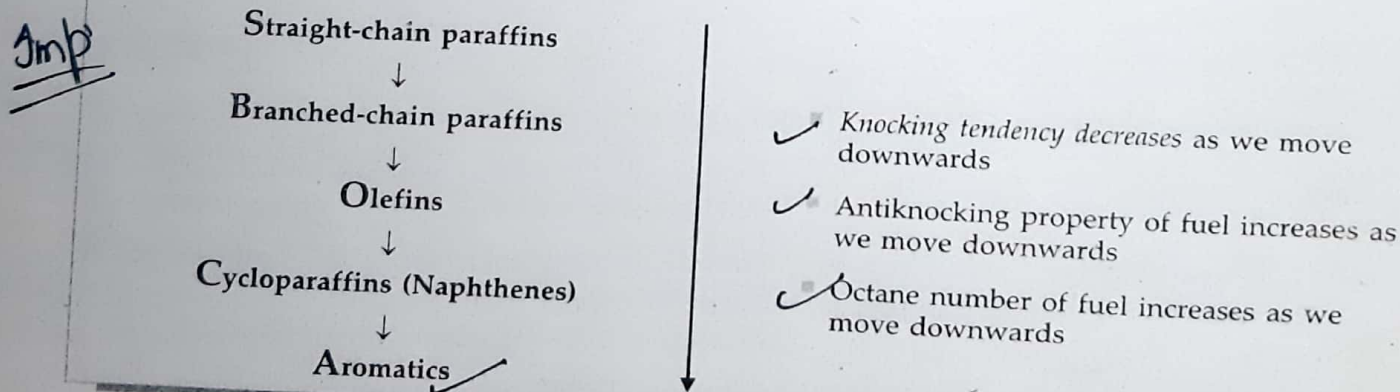
Furthermore, the anti-knocking tendency (and thus octane number) increases as the position of the double bond approaches the centre of chain.

(v) The knocking tendency decreases with increase in the compactness of the molecules. As a result, octane number of cyclo-paraffins (Naphthenes) are higher than n-alkanes, branched-alkanes and alkenes.

(vi) Aromatic hydrocarbons such as benzene and toluene have less knocking tendency than naphthenes so their octane numbers are highest.

Fuel	n-hexane	Cyclohexane	Benzene
Octane number	26	77	100 +

To sum up



To remember in reverse, use acronym : A C O B S

(vii) In cracked and reformed gasoline, it is the presence of olefins, iso-paraffins and aromatic hydrocarbons that increase its anti-knocking properties and thus the octane number.

NOTE Gasoline used in automobiles should have an octane number 80 or higher, while gasoline used in aeroplanes has an octane numbers 100 or even higher.

Engine	rpm	Fuel	Cetane number	Application
Low speed diesel engine	150 - 500	Low speed diesel fuel	> 25	Ships
Medium speed diesel engine	500 - 1500	Medium speed diesel fuel	> 35	Tractors or locomotives
High speed diesel engine	1500 - 2000	High speed diesel fuel	> 45	Trucks and buses

Additives for Improving Antiknock Properties

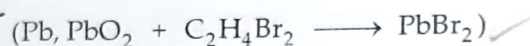
[GGSIPU, End Term, 2000]

Tetra ethyl lead (TEL) and diethyl telluride $\{(C_2H_5)_2Te\}$ are the most commonly used additives.

TEL gives rise to Pb and PbO during combustion. These particles act as free-radical chain inhibitors as they arrest the propagation of the explosive chain reactions responsible for knocking.

Efficiency of TEL decreases in presence of sulphur hence Desulphurized Gasoline is preferred.

Pb and PbO_2 decrease engine life hence they must be removed along with exhaust gases by adding ethylene dibromide.



Because $PbBr_2$ formed is volatile it escapes into atmosphere. But pollution problem still exists.

Another cause of pollution is incomplete combustion leading to the formation of CO , NO , NO_2 , SO_2 , SO_3 etc. Hence catalytic convertors based on Pt are employed which will catalyze combustion reaction to completion.

e.g.,



But Pt is poisoned by Pb , so Unleaded Petrol should be used.

Benzene is added for decreasing knocking. Since benzene is carcinogenic, very low concentration of benzene should be used.

10 UNLEADED PETROL

The petrol whose octane number is increased without the addition of lead compounds is referred to as unleaded petrol.

The Octane number of a fuel can be increased by :

(i) Reforming. This process increases the contents of molecules having branched and aromatic ring structures. Thus, by molecular reforming, the contents of isopentane, isooctane, ethyl benzene and isopropyl benzene increases in the given fuel.

(ii) Methyl tertiary butyl ether (MTBE), can also be added to improve the octane rating of the fuel. MTBE has oxygen in the form of ether group and supplies oxygen for the combustion of the petrol in internal combustion engine thus reducing the formation of peroxy-compounds.

Advantages of Using Unleaded Petrol

✓ Because of the absence of lead

- (i) unleaded petrol is free from lead-pollution.
- (ii) it allows the use of a catalytic converter to be attached to the exhaust in the automobiles.

NOTE Catalytic converter contains a rhodium catalyst which converts toxic gases like CO, NO etc. into harmless gases like CO_2 , N_2 etc. It also oxidises unburnt hydrocarbons into CO_2 and H_2O . Lead poisons the Rh catalyst by destroying the active sites.

11 DIESEL (OR COMPRESSION IGNITION) ENGINE AND CETANE NUMBER

[GGSIPU, IInd Term 2000, 02, 05 ; End Term 2000, 02, 06, 12]

In the diesel engine, air is first drawn into the cylinder and compressed to a pressure of about 500 psi ($3.52 \times 10^5 \text{ kg/m}^2$). This compression is accompanied by a rise in temperature to about 500°C .

Towards the end of the compression stroke, diesel is injected in the form of finely-divided spray into air in the cylinder heated to about 500°C by compression.

The oil absorbs the heat from the air and it ignites spontaneously as it attains ignition temperature. This raises the temperature and pressure. The piston is pushed by expanding gases in the power stroke.

The combustion of fuel in a diesel engine is not instantaneous, the interval between the start of fuel injection and its ignition is called the ignition delay.

Ignition delay is due to the time taken for the vaporization of the individual droplets and raising of the vapour to its ignition temperature.

It depends on the

- (a) Engine design ;
- (b) Efficiency of mixing of the spray and air
- (c) The injector design and
- (d) Mostly on the chemical nature of the fuel.

For example, ignition delay is shorter for paraffinic fuel, than for olefinic, naphthalenic and aromatic fuels.

If the ignition delay is long, it will lead to fuel accumulation in the engine even before the ignition. When ignited, an explosion results as the combined effect of increased temperature and pressure. This is responsible for diesel knock.

The diesel fuel should have a spontaneous ignition temperature less than the temperature produced by compression.

As the temperature to which air can be heated by compression is limited by various constraints, it is desirable to have fuels with short ignition delay, but the ignition delay must be long enough for the compression stroke to be completed.

In order to grade the diesel fuels, cetane rating is employed.

psi = pounds per square inch

1 psi = 6894.757 Pascals
1 atm = 14.7 psi

Cetane Nu

It is

It is

the same i

Cetane C
[or n - he

α -methy

C

F

ignite r

that hi

is fuel

and

alkyl r

less knock

with

and p

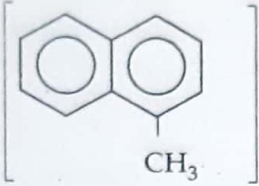
blen

Cetane Number

[GGSIPU, IInd Term, Nov. 2006]

It is a measure of the ease with which a fuel will ignite under compression. ✓

It is the % of hexadecane in a mixture of hexadecane and 1-methyl naphthalene which has the same ignition characteristics as the diesel fuel in question.

Diesel fuel	Cetane no.	Remarks
Cetane $C_{16}H_{34}$ [or n-hexadecane $CH_3(CH_2)_{14}CH_3$]	100	Very short ignition delay <i>so eliminates diesel knock.</i>
α -methyl naphthalene 	0	Longer ignition delay

Cetane no. of fuel primarily depends on the nature and composition of its hydrocarbons.

For instance, consider the following series :

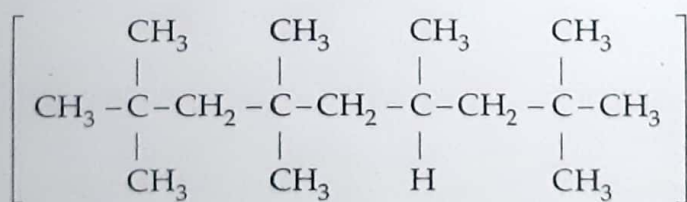
(Highest cetane number) n -alkanes > naphthenes (i.e., cycloalkanes) > alkenes > branched alkanes > aromatics (lowest cetane number)

- ↗ ignition delay increases from left to right
- ↗ ignition quality increases from right to left
- ↗ cetane no. increases from right to left.

As, straight chain alkanes like n -cetane have low ignition delay (high ignition quality) and ignite readily on compression, while aromatics do not ignite readily on compression, we can say that high cetane-no. fuels eliminate diesel knock. *Imp*

The cetane no. of diesel fuel may be raised by the addition of pre-ignition dopes such as alkyl nitrites such as ethyl nitrite, amyl nitrite etc.

2, 2, 4, 4, 6, 8, 8-hepta methyl nonane (HMN)



with a cetane rating of 15 is now considered as the low quality diesel in view of its easy availability and purity.

On the revised scale (HMN reference) the cetane number (CN) represents the % cetane in the blend with HMN plus $\frac{15}{100}$ of the % HMN.

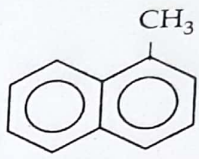
Thus, a blend of 50% cetane and 50% HMN has a cetane rating of

$$50 + \frac{15}{100} \times 50 = 57.5.$$

32 ENGINEERING CHEMISTRY (GGSIPU)

The differences between the Octane number and Cetane number are summarized below :

[GGSIPU, IInd Term 2003, 05, 10 ; End Term 2001, 11]

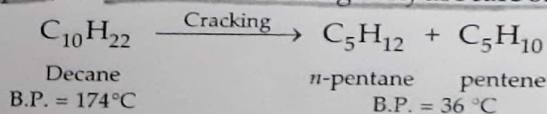
S.No.	Octane No.	Cetane No.
1.	It is the % of <i>iso</i>-octane in a mixture of isooctane (2, 2, 4-trimethyl pentane) and <i>n</i>-heptane, which matches the fuel under test in knocking characteristics. e.g., an 60-octane fuel is one which has the same combustion characteristics as a 60 : 40 mixture of <i>iso</i> octane and <i>n</i>-heptane. $\text{CH}_3 - (\text{CH}_2)_5 - \text{CH}_3 ;$ <i>n</i>-Heptane Knocks very-badly (Octane no. = 0) $\begin{array}{c} \text{CH}_3 \quad \quad \text{CH}_3 \\ \quad \quad \\ \text{CH}_3 - \text{C} - \text{CH}_2 - \text{C} - \text{CH}_3 \\ \quad \quad \\ \text{CH}_3 \quad \quad \text{CH}_3 \end{array}$ <i>iso</i>-octane or 2,2,4-Trimethyl Pentane Gives very little knowcking (Octane no. = 100)	It is the % of hexadecane in a mixture of hexadecane and 1-methyl naphthalene which has the same ignition characteristics as the diesel fuel in question.  1-methyl naphthalen (Cetane no. = 0) $\text{CH}_3 - (\text{CH}_2)_{14} - \text{CH}_3$ <i>n</i>-Hexadecane (Cetane no. = 100)
2.	It is used for finding suitability of gasoline or any other internal combustion engine fuel.	It is used for finding suitability of a diesel fuel.
3.	The Octane number of I.C. fuels can be raised by the addition of tetraethyl lead (C ₂ H ₅) ₄ Pb or TEL and Diethyl telluride (C ₂ H ₅) ₂ Te.	The cetane number of diesel fuel can be raised by the addition of small quantity of certain 'Pre-ignition dopes' like ethyl nitrite, isoamyl nitrite, acetone peroxide etc.
4.	Hydrocarbons which are poor diesel fuels are quite good gasoline fuels.	Ignition quality order among hydrocarbon constituent of a diesel fuel is as follows : <i>n</i> -alkanes > naphthalenes > alkenes > branched alkanes > aromatics. Thus, hydrocarbons which are poor gasoline fuels are quite good diesel fuels.
5.	Straight chain hydrocarbon molecules are worst fuels. As they have low-Octane numbers.	Straight chain hydrocarbon molecules are best, as they have high cetane numbers.

12 CRACKING

at high temp.

[GGSIPU, IInd Term, Dec. 2001, Nov. 2007]

Cracking is defined as the decomposition of high molecular weight hydrocarbons of high boiling points into simpler, lower molecular weight hydrocarbons of low boiling points.

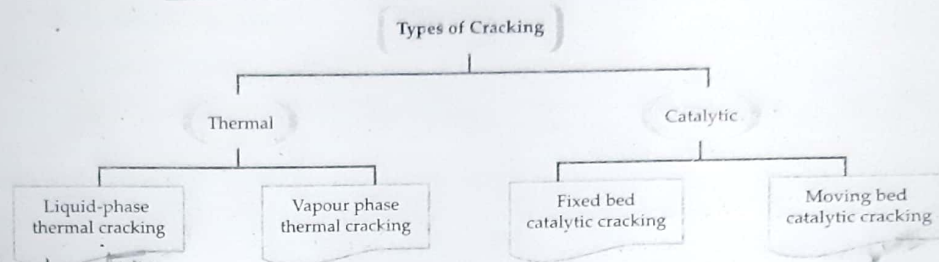


Importance. Of all the fractions obtained by fractionation of petroleum, gasoline has the highest demand as a motor fuel, but the yield of this fraction is only 20% of the crude.

Also the quality of so-called 'straight-run' gasoline is not so good.

Hence, it is used only after suitable blending. Moreover, there is a surplus of heavier petroleum fractions. To overcome these difficulties, the middle and heavy fractions are cracked to get more valuable low boiling fractions.

For instance, the petrol made by cracking has far better characteristics (as far as the IC engine is concerned) than 'straight-run' petrol.



12.1 Thermal Cracking

[GGSIPU, 2005]

When the heavy oils are subjected to high temperature and pressure in the absence of catalyst, it is called thermal cracking. In a thermal cracking process, the heavy oil residue (from primary distillation of crude oil) : (a) is heated to a temperature greater than its boiling point to promote rupture of the carbon-carbon bonds and (b) subjected to high pressures to prevent the vaporization of the cracked residues.

In thermal cracking the bigger hydrocarbon molecules break down to give smaller molecules of the paraffin, olefins plus some hydrogen. It is of two types viz. liquid and vapour phase thermal cracking. The operating conditions depend on the desired product. Low pressures and high temperature (that is conditions for vapour phase thermal cracking) tend to produce more low-molecular weight and volatile hydrocarbons than those produced under high pressure and low temperature conditions (that is under liquid phase thermal cracking). A comparative account of these two is given in Table 7.

Table 7 Comparison of Liquid and Vapour phase Thermal Cracking

[GGSIPU, IInd Term 2003]

Characteristic	Liquid-phase-Thermal Cracking	Vapour Phase-Thermal Cracking
Cracking temperature	475 – 530 °C	600 – 650 °C
Pressure	100 kg / m sec ²	10 – 20 kg / m sec ²
Yield of Petrol	50 – 60%	—
Octane rating of Petrol	65 – 70	(> 70) ∴ Better antiknock characteristics
Criteria	None Any heavy oil can be used	Oil should be readily vaporized.
Time required for cracking	More	Less

NOTE

$$P = \frac{F}{A} = \frac{m \cdot a}{A} = \frac{\text{kg}}{\text{m}^2} \cdot \frac{\text{m}}{\text{sec}^2} = \frac{\text{kg}}{\text{m sec}^2}$$

Thermal cracking products = Paraffin, Olefins, H₂

Condition Liquid phase
 • High pressure and low temp.
 • Produce less Volatile Hydrocarbons

Vapour
 • High Temp. and Low pressure
 • produce Low mole. weight and volatile

Fixed Bed

12.2 Catalytic Cracking

Moving Bed

[GGSIPU, IInd Term 2005 ; End Term, 2010]

- property of gasoline is improved
- Low temp. is required
- Low press. is required
- Most recent catalyst = zeolite.

The quality and quantity of gasoline produced by cracking can be greatly improved by using a suitable catalyst like aluminium silicate, $Al_2(SiO_2)_3$ or Synthetic, hydrated aluminosilicates ($Al_2O_3 = 12.5\%$ and $SiO_2 = 87.5\%$) or acid-activated natural silicates like montmorillonites or molecular sieves. Catalytic cracking requires much lower temperatures and pressures compared to thermal cracking. There are two types of catalytic cracking, viz. Fixed bed and moving bed catalytic cracking. These are discussed below :

12.2.1 Fixed-bed Catalytic Cracking

[GGSIPU, IInd Term 2005]

The heavy oil charge is passed through a preheater where it is heated upto $425 - 450^\circ C$. The hot vapours are then passed over a fixed bed of catalyst chamber maintained at about $425 - 450^\circ C$ and 1.5 kg/cm^2 pressure.

During their passage through the tower, about 40% of the charge is converted into gasoline and about 2 - 4% C is formed. The latter gets adsorbed on the catalyst bed. The catalyst, after 8 to 10 hours, stops functioning, due to the deposition of black layer of carbon formed during cracking. This is re-activated by burning off the deposited carbon. During the re-activation interval, the vapours are diverted through another catalyst chamber.

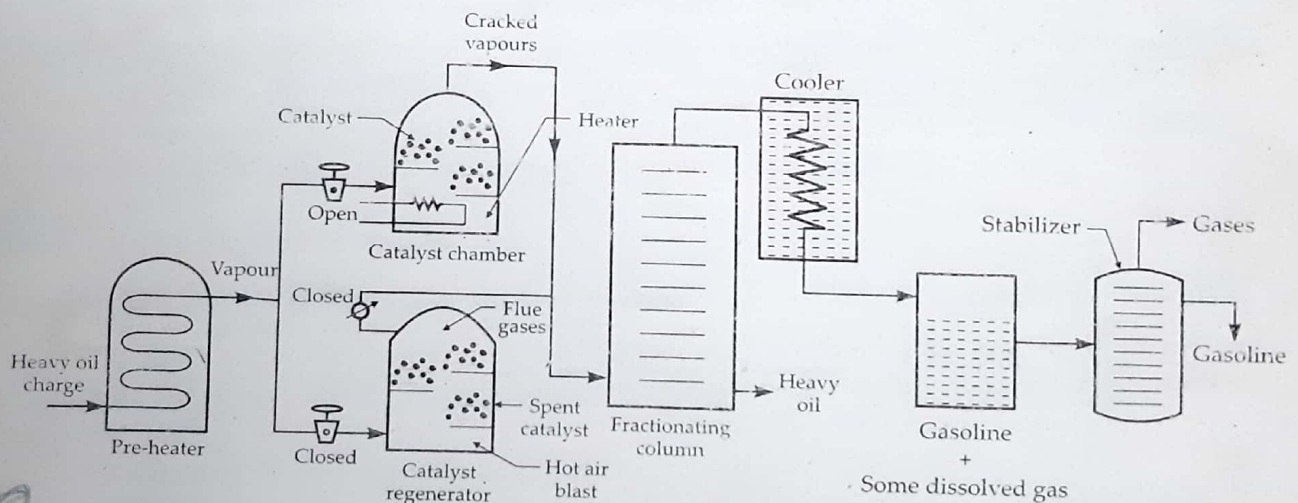


Figure 5 Fixed bed catalytic cracking process.

12.2.2 Moving-bed or Fluidized-bed Catalytic Cracking

[GGSIPU, IInd Term 2005]

In moving bed catalytic cracking, the feed oil is first passed through a pre-heater. The pre-heated oil vapours along with very finely powdered catalyst are then passed in a reactor which is maintained at a temperature of $500^\circ C$ for catalytic cracking. The cracked oil vapours are then passed to the fractionating column where heavy oil is separated. The vapours are then passed through the cooler where gasoline condenses along with some gases. This is then sent to a stabiliser where the dissolved gases are removed and pure gasoline is recovered. The main components along with their functions are discussed with the help of Fig. 6.

Vapours + catalyst

Heater

Cracked

Fractional

column

(Heavy oil gets separated)

Vapours feed to cooler

gasoline and gases condensed

Pass to

pure gasoline

stabilizer

Advantages of Catalytic Cracking over Thermal cracking

[GGSIPU, IInd Term 2005]

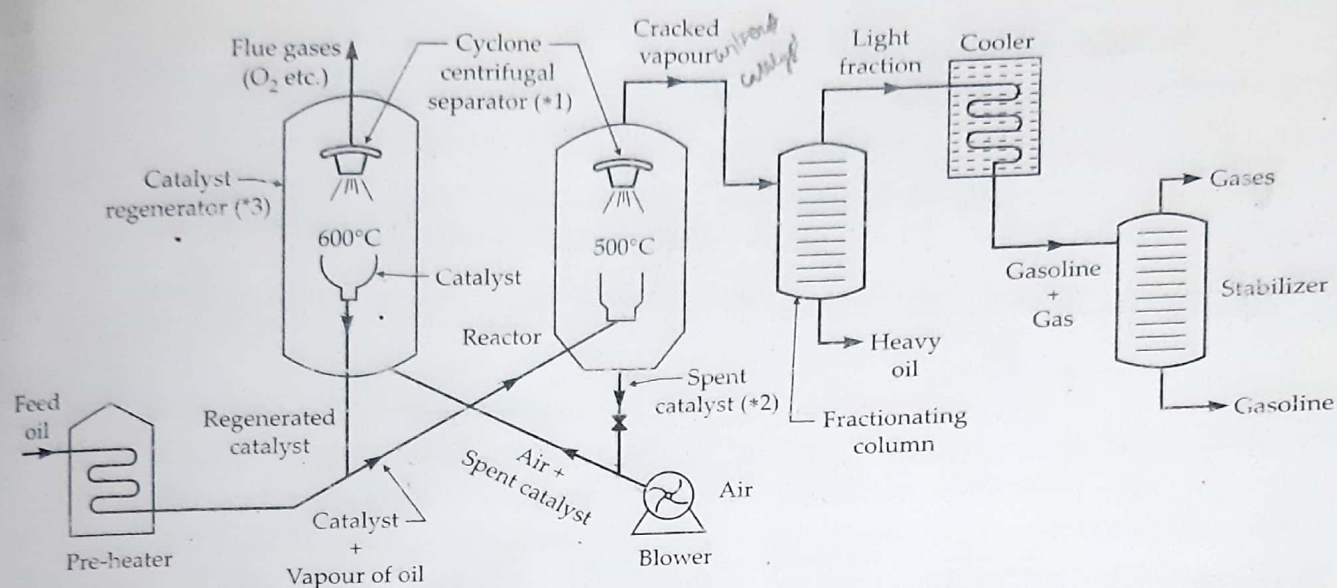
✓ 1. The yield of petrol is higher, because catalysts are selective in their action and therefore, they permit the cracking of only the high-boiling hydrocarbons. Thus, the evolution of by-product gases can be minimized.

2. The quality of petrol produced is better

- ✓ (i) The product of cracking contains a higher amount of aromatics and hence, it possess better anti-knock characteristic.
- (ii) Isomerization to branched-chain compounds (iso-paraffins) occur, thereby better petrol is produced.
- (iii) The product contains a lesser amount of undesirable sulphur, because a major portion of it escapes as H_2S gas, during cracking.
- (iv) Decomposition of aromatic removes only the side chains, but no ring itself is broken.

NOTE Tendency of fuel-constituents to knock : straight chain paraffins > branched-chains paraffins > olefins > naphthalenes > aromatics.

3. No external fuel is necessary for cracking. The heat required for cracking is derived from the coal, embedded in the catalyst.



- * 1 Cyclone allows only the cracked oil vapours to pass on to the 'fractionating column', but retains all the catalyst powder in the reactor itself.
- * 2 The catalyst powder gradually becomes heavier, due to coating with carbon, and settles to the bottom, from where it is forced by an air blast to regenerator (maintained at 600°C).
- * 3 In regenerator, carbon is burnt and the regenerated catalyst then flows through stand-pipe for mixing with fresh batch of incoming cracking oil.

Figure 6 Moving bed catalytic cracking.

Advantages of Catalytic Cracking over Thermal cracking

[GGSIU, IInd Term 2005]

1. The yield of petrol is higher, because catalysts are selective in their action and therefore, they permit the cracking of only the high-boiling hydrocarbons. Thus, the evolution of by-product gases can be minimized.

2. The quality of petrol produced is better

- (i) The product of cracking contains a higher amount of aromatics and hence, it possess better anti-knock characteristic.
- (ii) Isomerization to branched-chain compounds (iso-paraffins) occur, thereby better petrol is produced.
- (iii) The product contains a lesser amount of undesirable sulphur, because a major portion of it escapes as H₂S gas, during cracking.
- (iv) Decomposition of aromatic removes only the side chains, but no ring itself is broken.

NOTE Tendency of fuel-constituents to knock : straight chain paraffins > branched-chains paraffins > olefins > naphthalenes > aromatics.

3. No external fuel is necessary for cracking. The heat required for cracking is derived from the coal, embedded in the catalyst.

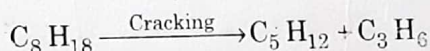
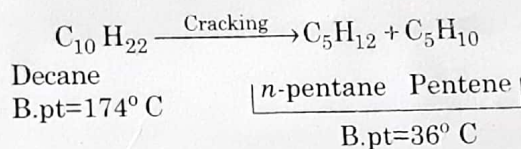
with the help of cooling coils. For example when crude oil is made to pass through first column, the temperature of which is so maintained that fuel oil condense at the bottom and the vapour of other components are passed on to second fractionating column. In the second column, the temperature is so maintained that vapours of gas oil condense at the bottom and so on. Steam is injected at the bottom of each fractionating column for its partial pressure effect. Vapour of gasoline finally pass out from the top (Fig. 1.13).

In top flashing, there is better control of product composition, but requires more pumps and instruments and hence is an expensive process.

1.5.5 Cracking

Of all the fractions obtained by fractional distillation of crude petroleum, gasoline is considered to be the most important fraction. With the rapid growth of automobile industry, the consumption of petrol has increased many folds, but the yield of this fraction is only 20% of the crude oil. Also the quality of the gasoline obtained is not good. This gasoline is called straight run gasoline. On the other hand, the yield of heavier petroleum fractions is quite high. Thus, it is necessary to make better use of these heavy fractions, by converting them into more useful fraction, gasoline. This is achieved by a technique called 'cracking'.

'Cracking' is the process by which heavier fractions are converted into lighter fractions by the application of heat, with or without catalyst. Cracking involves the rupture of C—C and C—H bonds in the chains of high molecular weight hydrocarbons. For example-



Nearly 50% of today's gasoline is obtained by cracking. It may be emphasized here, that the gasoline obtained by cracking is far more superior to the straight run gasoline. The process of cracking involves the following chemical changes:

- Higher hydrocarbons are converted to lower hydrocarbons by C—C cleavage. The products obtained on cracking have low boiling points than initial reactant.
- Formation of branched chain hydrocarbons takes place from straight chain alkanes.
- Unsaturated hydrocarbons are formed from saturated hydrocarbons.
- Cyclization may take place.

Though, the chief application of commercial cracking is the production of gasolines but it is also used for the production of olefins from naphthas, oil gas from kerosene and the reduction of viscosity of straight run residues. Cracking can be carried out by two methods:

- Thermal Cracking:** When cracking takes place simply by the application of heat and pressure, the process is called thermal cracking. The heavy oils are subjected to high temperature and pressure, when the bigger hydrocarbons break down to give smaller molecules of paraffins, olefins etc. The thermal stability among the constituents of petroleum fractions increases as Paraffins < naphthenes (alicyclics) < aromatics. The products obtained, depend upon the operating conditions. Thermal cracking may be carried out either in liquid phase or vapour phase.

- Liquid phase thermal cracking:** By this method, any type of oil (residues, heavy oil) can be cracked. The charge is kept in the liquid form by applying high pressures of

By applying
Heat and
Pressure
without
catalyst.
Liq.
Vapour.

poor quality
straight run gasoline
↑
Lower yield of gasoline
Higher yield of other heavy fractions
poorly.

the range 30 kg/cm² to 100 kg/cm² at a suitable temperature of 476–530° C. The cracked products are separated in a fractionating column. The important fractions are: Cracked gasoline (30-35%); Cracking gases (10-45%); Cracked fuel oil (50-55%).

The cracked gasoline has an octane number of about 70. The cracking gases constitute raw materials for petrochemicals. The cracking residue is used as a fuel.

- (b) Vapour phase thermal cracking: By this method, only those oils which can vaporize at low temperatures can be cracked. The petroleum fractions of low boiling range like kerosene oil, are heated at a temperature of 670-720° C under low pressure of 10-20 kg/cm². The time required for cracking in this process is much less as compared to liquid phase thermal cracking. Petrol obtained from vapour phase cracking has better antiknock properties but poorer stability.

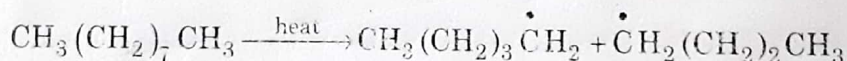
In general, thermal cracking produces petrols which are less stable than straight run petrols due to the presence of highly reactive unsaturated hydrocarbons in them.

Table 1.6 Comparison of liquid and vapour phase thermal cracking.

Characteristic	Liquid Phase-Thermal Cracking	Vapour Phase-Thermal Cracking
Cracking temperature	475 – 530°C	670 – 720°C
Pressure	100 kg/cm ²	10 – 20 kg/cm ²
Octane rating of Petrol	65 – 70	> 70 ✓
Type of oil	Any heavy oil can be cracked	Oil should be readily vapourized

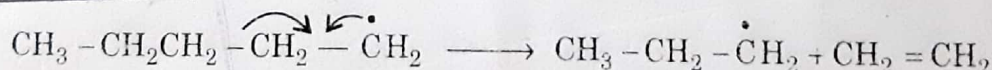
Mechanism of thermal cracking: Thermal cracking is believed to follow free radical mechanism. The mechanism can be illustrated by taking the example of n-nonane.

Initiation

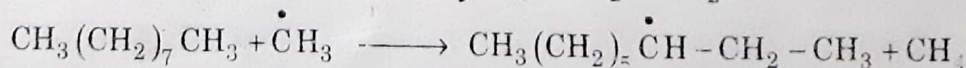
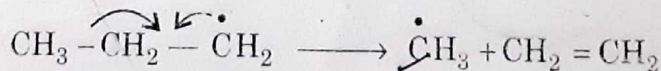


Propagation

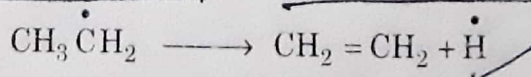
The free radicals formed are thermally unstable and undergo fission at the β-position to yield a new radical and an olefin.



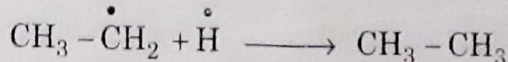
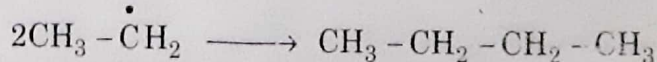
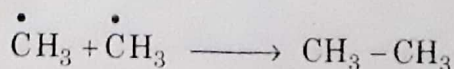
This β scission continues, till a thermally more stable radical is formed.



The radicals having no C—C bond in the β position give rise to hydrogen free radicals.



Termination



Catalytic Cracking: In this method the cracking is brought about in the presence of a catalyst at much lower temperatures and pressures. The catalyst used is usually a mixture of silica and alumina (hydrated aluminium silicate). The most recent catalyst used is zeolite. The quality and yield of gasoline produced by catalytic cracking is greatly improved. The catalytic cracking has following advantages over the thermal cracking :

- (i) High temperatures and pressures are not required in presence of a catalyst and hence there is saving in the production costs.
- (ii) The use of catalyst not only accelerates the cracking reactions (decomposition, dehydrogenation, isomerization and polymerization), but also introduces new reactions which considerably modify the yield and the nature of the products.
- (iii) The yield of the gasoline is higher. About 70% of raw material is converted into gasoline.
- (iv) The gasoline obtained by catalytic cracking contains low percentage of olefins and high percentage of isoparaffin and aromatic hydrocarbons. This increases the antiknock properties of gasoline. Gasoline formed by catalytic cracking has an octane number of 80.
- (v) No external fuel is required for cracking, since the heat required is obtained by burning off the coke deposited on the catalyst itself, during regeneration process.
- (vi) The process can be better controlled than that of thermal process, so desired products can be obtained.
- (vii) The product contains a very little amount of undesirable sulphur, because a major portion of it escapes out as H_2S gas, during cracking.
- (viii) The gasoline formed contains much less gum and gum forming compounds.
- (ix) Compared with the thermal process, the catalytic cracking yields less coke, less gas and more liquid products.
- (x) The evolution of by-product gas can be further minimized, thereby increasing the yield of desired petrol.
- (xi) Catalysts are selective in action and hence cracking of only high boiling fractions takes place.
- (xii) Coke forming materials are absorbed by the catalysts as soon as they are formed.

Depending upon how the catalyst is used, the catalyst cracking is of two types:

- (i) **Fixed bed catalytic Cracking:** In this type of cracking the catalyst is in the form of granules or pellets. These pellets or granules are used in the form of fixed beds in catalyst towers.

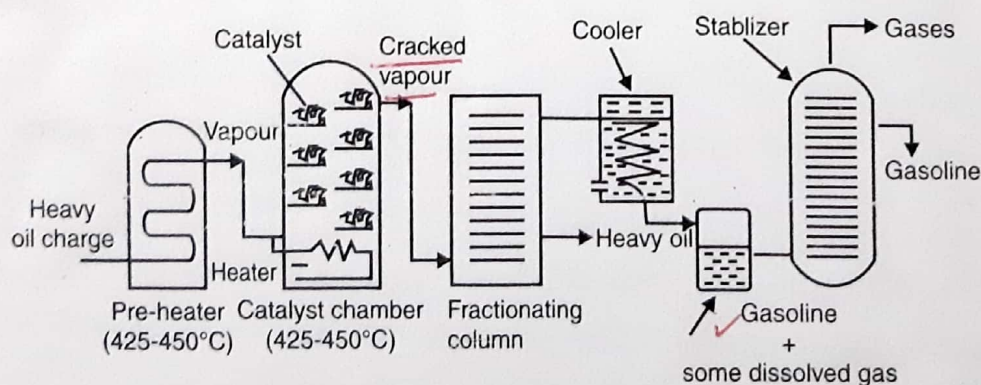


Fig. 1.14 Fixed-bed type catalytic cracking

the range 30 kg/cm^2 to 100 kg/cm^2 at a suitable temperature of $476-530^\circ \text{C}$. The cracked products are separated in a fractionating column. The important fractions are: Cracked gasoline (30-35%); Cracking gases (10-45%); Cracked fuel oil (50-55%).

The cracked gasoline has an octane number of about 70. The cracking gases constitute raw materials for petrochemicals. The cracking residue is used as a fuel.

- (b) Vapour phase thermal cracking: By this method, only those oils which can vaporize at low temperatures can be cracked. The petroleum fractions of low boiling range like kerosene oil, are heated at a temperature of $670-720^\circ \text{C}$ under low pressure of $10-20 \text{ kg/cm}^2$. The time required for cracking in this process is much less as compared to liquid phase thermal cracking. Petrol obtained from vapour phase cracking has better antiknock properties but poorer stability.

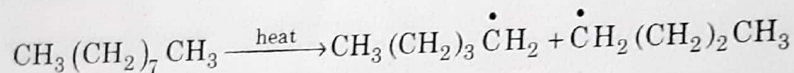
In general, thermal cracking produces petrols which are less stable than straight run petrols due to the presence of highly reactive unsaturated hydrocarbons in them.

Table 1.6 Comparison of liquid and vapour phase thermal cracking.

Characteristic	Liquid Phase-Thermal Cracking	Vapour Phase-Thermal Cracking
Cracking temperature	$475 - 530^\circ \text{C}$	$670 - 720^\circ \text{C}$
Pressure	100 kg/cm^2	$10 - 20 \text{ kg/cm}^2$
Octane rating of Petrol	65 - 70	> 70
Type of oil	Any heavy oil can be cracked	Oil should be readily vapourized

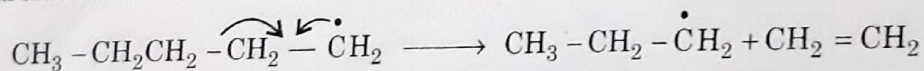
Mechanism of thermal cracking: Thermal cracking is believed to follow free radical mechanism. The mechanism can be illustrated by taking the example of n-nonane.

Initiation

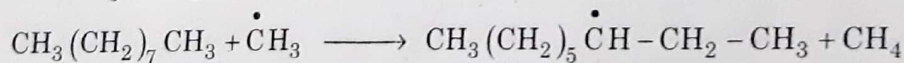
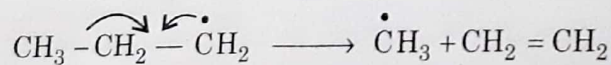


Propagation

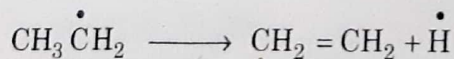
The free radicals formed are thermally unstable and undergo fission at the β -position to yield a new radical and an olefin.



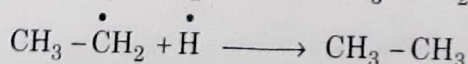
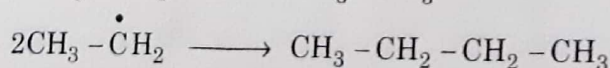
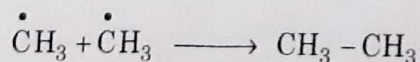
This β scission continues, till a thermally more stable radical is formed.



The radicals having no C—C bond in the β position give rise to hydrogen free radicals.



Termination



2. **Catalytic Cracking:** In this method the cracking is brought about in the presence of a catalyst at much lower temperatures and pressures. The catalyst used is usually a mixture of silica and alumina (hydrated aluminium silicate). The most recent catalyst used is zeolite. The quality and yield of gasoline produced by catalytic cracking is greatly improved. **The catalytic cracking has following advantages over the thermal cracking :**

- (i) High temperatures and pressures are not required in presence of a catalyst and hence there is saving in the production costs.
- (ii) The use of catalyst not only accelerates the cracking reactions (decomposition, dehydrogenation, isomerization and polymerization), but also introduces new reactions which considerably modify the yield and the nature of the products.
- (iii) The yield of the gasoline is higher. About 70% of raw material is converted into gasoline.
- (iv) The gasoline obtained by catalytic cracking contains low percentage of olefins and high percentage of isoparaffin and aromatic hydrocarbons. This increases the antiknock properties of gasoline. Gasoline formed by catalytic cracking has an octane number of 80.
- (v) No external fuel is required for cracking, since the heat required is obtained by burning off the coke deposited on the catalyst itself, during regeneration process.
- (vi) The process can be better controlled than that of thermal process, so desired products can be obtained.
- (vii) The product contains a very little amount of undesirable sulphur, because a major portion of it escapes out as H_2S gas, during cracking.
- (viii) The gasoline formed contains much less gum and gum forming compounds.
- (ix) Compared with the thermal process, the catalytic cracking yields less coke, less gas and more liquid products.
- (x) The evolution of by-product gas can be further minimized, thereby increasing the yield of desired petrol.
- (xi) Catalysts are selective in action and hence cracking of only high boiling fractions takes place.
- (xii) Coke forming materials are absorbed by the catalysts as soon as they are formed.

Depending upon how the catalyst is used, the catalytic cracking is of two types:

- (i) **Fixed bed catalytic Cracking:** In this type of cracking the catalyst is in the form of granules or pellets. These pellets or granules are used in the form of fixed beds in catalyst towers.

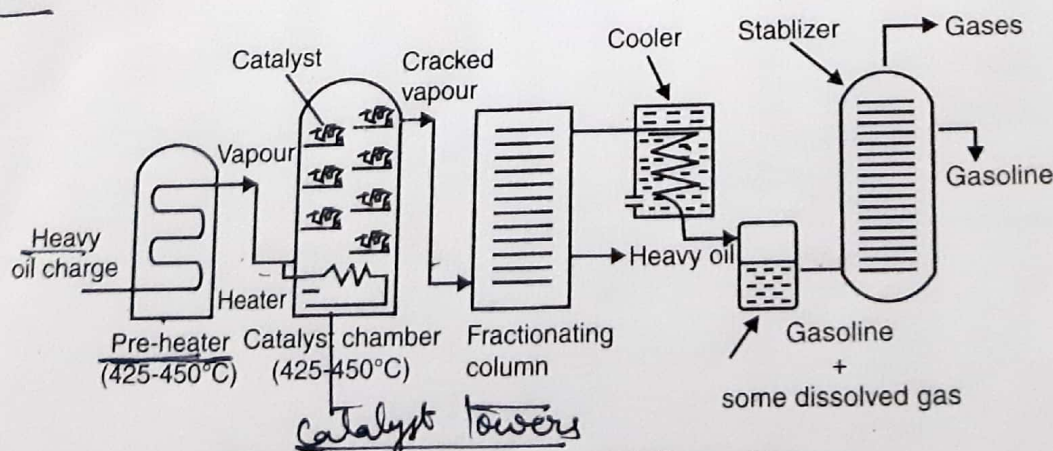


Fig. 1.14 Fixed-bed type catalytic cracking

The heavy oil charge is passed through a heater and heated to a cracking temperature (420-450°C). The charge is then passed through a catalyst chamber. The catalyst used is either silica-alumina gel ($\text{SiO}_2 \cdot \text{Al}_2\text{O}_3$) or bauxite, mixed with clay and zirconium oxide. The catalyst tower is also maintained at 420-450°C and at a pressure of 1 to 5 kg/cm². Cracking of the heavy oil vapours takes place in the catalytic chamber and about 30-40% of the charge is converted into low molecular weight hydrocarbons. The carbon formed (about 4%) during the cracking process gets adsorbed on the catalyst thereby deactivating it. The cracked vapours are passed through the fractionating column, where the gasoline vapours are recovered from the top and the heavy oil gets condensed at the bottom. The vapours are recovered from the top and the heavy oil gets condensed at the bottom. The gasoline vapours are subjected to a cooler where gasoline along with some other gaseous products gets condensed. Thereafter the gasoline is sent to a stabilizer, where the dissolved gases are removed from the top and pure gasoline is obtained, having the desired boiling range (Fig. 1.14).

During the process, the catalyst stops working after about 10-11 hours, due to the deposition of carbon particles formed during the cracking process. This is reactivated by burning off the carbon in a stream of hot air. During the reactivation of the catalyst, the vapours of heavy oil are diverted through another stand by catalytic chamber so that the reaction proceeds continuously.

As the cracking reactions are endothermic, the temperature in the catalytic chamber is maintained by circulating molten salt (mixture of sodium nitrate and sodium nitrite) in pipes passing through the catalyst bed.

- (ii) **Moving bed Catalytic Cracking:** This is also known as fluid bed catalytic cracking. In this type of cracking, the catalyst is in the form of fine powder which flows down to the cracking chamber, through a hopper. The catalyst gets mixed with the heavy oil vapours and behave much like a fluid which can be circulated in gas streams. The plant for fluid bed or moving bed catalyst cracking consists of a reactor and a regenerator placed side by side (Fig. 1.15)

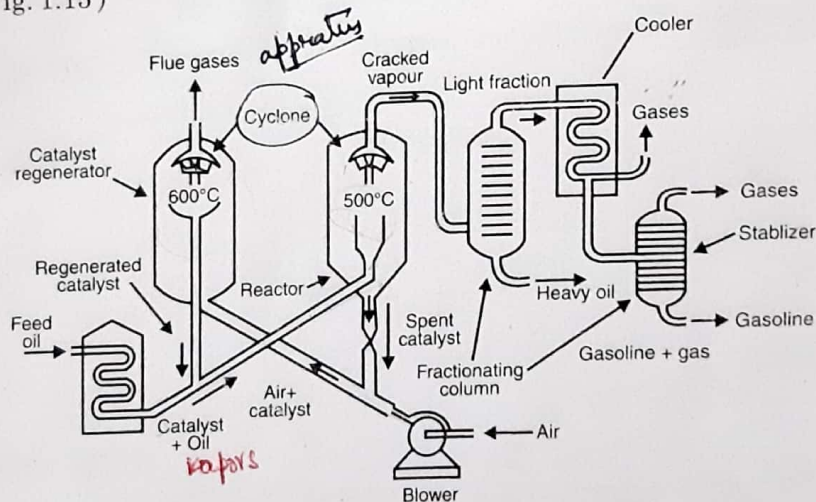
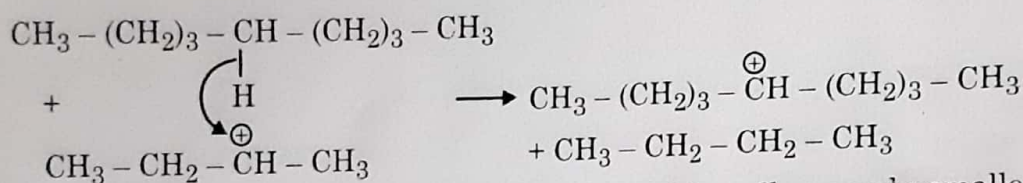
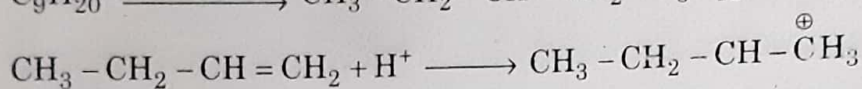


Fig. 1.15 Moving-bed type catalytic cracking

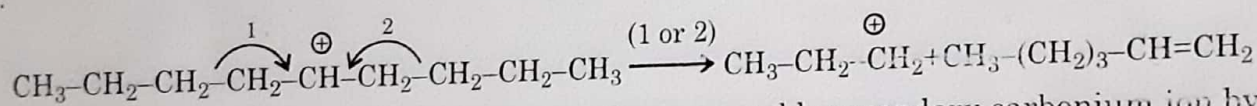
The heavy oil is heated in a heater and the vapours are allowed to mix with the hot catalyst, in the form of fine powder, which is introduced from the regenerator. The mixture is forced into a large reactor maintained at 500°C. Due to the high temperature of the catalyst (600°C), the cracking of the heavy oil starts even before its entry into the reactor. This facilitates the cracking

Mechanism: Catalytic cracking is believed to involve carbonium-ion intermediates. This can be illustrated by taking the example of n-nonane. First, due to thermal cracking, small amount of alkenes is produced. The alkene produced is thus protonated by the catalyst.

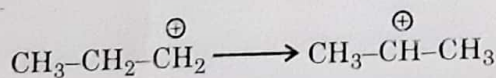
Mechanism: Catalytic cracking is believed to involve carbonium-ion intermediates. This can be illustrated by taking the example of n-nonane. First, due to thermal cracking, small amount of alkenes is produced. The alkene produced is thus protonated by the catalyst.



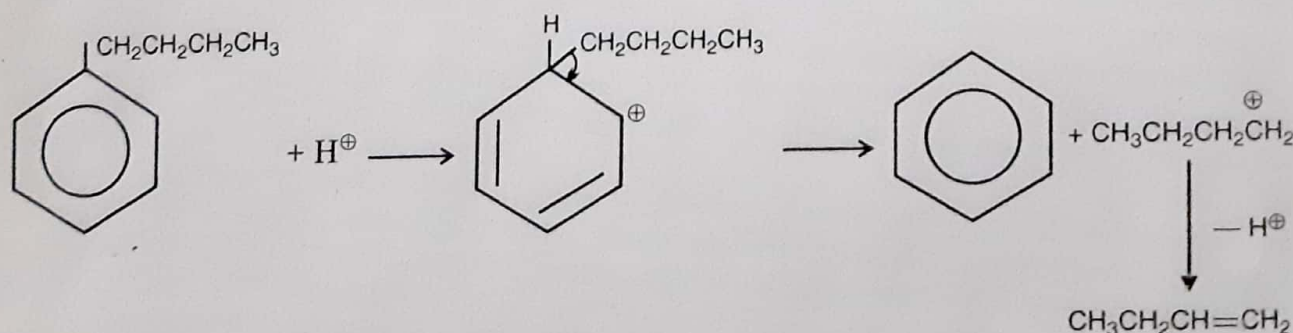
The nonyl carbonium ion then undergoes β cleavage to yield an alkene and a smaller carbonium ion.



The primary carbonium ion rearrange to the more stable secondary carbonium ion by 1,2 hydride shift.



In aromatic hydrocarbons, the rings remain intact and only the side chains are lost.



Factors on which knocking depends :

- (i) Engine design (ii) Running conditions and
(iii) Chemical structure of the fuel hydrocarbons. For instance :

knocking tendency decreases in the order —

n -alkanes > mono-substituted alkanes > cyclo alkanes > alkenes
> poly substituted alkenes > aromatics.

And for straight chain hydrocarbons, the tendency to knock increases with molecular weight and boiling point, e.g.,

n -hexane > n -pentane > n -butane.

Aromatic hydrocarbons have higher antiknocking properties than paraffins and olefins.

9 OCTANE NUMBER

Knocking Property of the Fuel is expressed in Octane number.
[GGSIPU, IInd Term 2000, 02, 05 ; End Term 2000, 02, 06, 12]

The resistance offered by gasoline to knocking cannot be defined in absolute terms. It is generally expressed on an arbitrary scale, known as **Octane rating** (proposed by Graham Edgar in 1926).

(More knocking) straight chain

Fuel	Octane number	Characteristics
n -heptane	0	Knocks severely
iso-octane	100	High resistance to knocking

chain →

The % of iso-octane in the n -heptane — iso-octane blend which has the same knocking characteristics as the gasoline sample, under the same set of conditions is known as octane number.

9.1 Influence of Structure of the Fuel on the Octane Rating

[GGSIPU, IInd Term 2002 ; End Term 2000, 2006]

(i) As straight chain (or normal) hydrocarbons knock severely in petrol engines so they have lowest octane numbers.

(ii) As the knocking tendency of n -alkanes increases with the increase in the length of carbon chain so anti-knocking properties (and thus octane numbers) also decrease with increase in the length of the carbon chain.

For example :

n -paraffin	Octane number
Methane	122
Ethane	101
Propane	96
n -butane	89
n -pentane	60
n -hexane	29
n -heptane	0
n -nonane	-45

In the same engine under the same set of conditions and the same critical CR, various blends of n -heptane and iso-octane are burnt and % of iso-octane by volume in blend that knocks under these conditions is octane number of gasoline.

More octane
no less the
knocking

Petrol

Leaded

Unleaded

Incomplete

↑ efficiency

combustion

so ↓ efficiency.

(iii) Branched chain paraffins have lower tendency to knock so they have higher anti-knock properties (and thus octane numbers) in comparison to normal chain paraffins.

Furthermore, resistance to knock increases with the number of branches and their position.

For example :

Hydrocarbon	n-heptane	2-methyl hexane	2, 2-dimethyl pentane
Octane number	0	55	80 ✓

Recently, the hydrocarbon 2,2,3-trimethylpentane called triptane has been assigned an octane number, 125.

(iv) Knocking tendency decreases with the increase in the number of carbon-carbon double bonds in the hydrocarbon fuel.

Thus, olefins have higher anti-knock properties (and thus octane number) than the paraffins of same carbon-chain length.

Furthermore, the anti-knocking tendency (and thus octane number) increases as the position of the double bond approaches the centre of chain.

(v) The knocking tendency decreases with increase in the compactness of the molecules. As a result, octane number of cyclo-paraffins (Naphthenes) are higher than n-alkanes, branched-alkanes and alkenes.

(vi) Aromatic hydrocarbons such as benzene and toluene have less knocking tendency than naphthenes so their octane numbers are highest.

Fuel	n-hexane	Cyclohexane	Benzene
Octane number	26	77	100 +

To sum up

Straight-chain paraffins
 ↓
 Branched-chain paraffins
 ↓
 Olefins
 ↓
 Cycloparaffins (Naphthenes)
 ↓
 Aromatics ✓

- ✓ Knocking tendency decreases as we move downwards
- ✓ Antiknocking property of fuel increases as we move downwards
- ✓ Octane number of fuel increases as we move downwards

To remember in reverse, use acronym : A C O B S

(vii) In cracked and reformed gasoline, it is the presence of olefins, iso-paraffins and aromatic hydrocarbons that increase its anti-knocking properties and thus the octane number.

NOTE Gasoline used in automobiles should have an octane number 80 or higher, while gasoline used in aeroplanes has an octane numbers 100 or even higher.

Engine	rpm	Fuel	Cetane number	Application
Low speed diesel engine	150 – 500	Low speed diesel fuel	> 25	Ships
Medium speed diesel engine	500 – 1500	Medium speed diesel fuel	> 35	Tractors or locomotives
High speed diesel engine	1500 – 2000	High speed diesel fuel	> 45	Trucks and buses

Additives for Improving Antiknock Properties

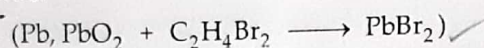
[GGSIPU, End Term, 2000]

Tetra ethyl lead (TEL) and diethyl telluride $\{(C_2H_5)_2Te\}$ are the most commonly used additives.

TEL gives rise to Pb and PbO during combustion. These particles act as free-radical chain inhibitors as they arrest the propagation of the explosive chain reactions responsible for knocking.

Efficiency of TEL decreases in presence of sulphur hence Desulphurized Gasoline is preferred.

Pb and PbO_2 decrease engine life hence they must be removed along with exhaust gases by adding ethylene dibromide.



Because $PbBr_2$ formed is volatile it escapes into atmosphere. But pollution problem still exists.

Another cause of pollution is incomplete combustion leading to the formation of CO, NO, NO_2, SO_2, SO_3 etc. Hence catalytic convertors based on Pt are employed which will catalyze combustion reaction to completion.



But Pt is poisoned by Pb , so Unleaded Petrol should be used.

Benzene is added for decreasing knocking. Since benzene is carcinogenic, very low concentration of benzene should be used.

10 UNLEADED PETROL

The petrol whose octane number is increased without the addition of lead compounds is referred to as unleaded petrol.

The Octane number of a fuel can be increased by :

(i) **Reforming.** This process increases the contents of molecules having branched and aromatic ring structures. Thus, by molecular reforming, the contents of isopentane, isooctane, ethyl benzene and isopropyl benzene increases in the given fuel.

(ii) **Methyl tertiary butyl ether (MTBE).** can also be added to improve the octane rating of the fuel. MTBE has oxygen in the form of ether group and supplies oxygen for the combustion of the petrol in internal combustion engine thus reducing the formation of peroxy compounds.

Advantages of Using Unleaded Petrol

✓ Because of the absence of lead

- unleaded petrol is free from lead-pollution.
- it allows the use of a catalytic converter to be attached to the exhaust in the automobiles.

NOTE Catalytic converter contains a rhodium catalyst which converts toxic gases like CO, NO etc. into harmless gases like CO_2 , N_2 etc. It also oxidises unburnt hydrocarbons into CO_2 and H_2O . Lead poisons the Rh catalyst by destroying the active sites.

11 DIESEL (OR COMPRESSION IGNITION) ENGINE AND CETANE NUMBER

[GGSIPU, IInd Term 2000, 02, 05 ; End Term 2000, 02, 06, 12]

In the diesel engine, air is first drawn into the cylinder and compressed to a pressure of about 500 psi ($3.52 \times 10^5 \text{ kg/m}^2$). This compression is accompanied by a rise in temperature to about 500°C . *Imp*

Towards the end of the compression stroke, diesel is injected in the form of finely-divided spray into air in the cylinder heated to about 500°C by compression.

The oil absorbs the heat from the air and it ignites spontaneously as it attains ignition temperature. This raises the temperature and pressure. The piston is pushed by expanding gases in the power stroke.

The combustion of fuel in a diesel engine is not instantaneous, the interval between the start of fuel injection and its ignition is called 'the ignition delay'. *Imp* Lesser the ignition delay

Ignition delay is due to the time taken for the vaporization of the individual droplets and raising of the vapour to its ignition temperature.

It depends on the

- Engine design ;
- Efficiency of mixing of the spray and air
- The injector design and
- Mostly on the chemical nature of the fuel.

For example, ignition delay is shorter for paraffinic fuel, than for olefinic, naphthalenic and aromatic fuels. *(Alkanes)* *(Alkenes)*

If the ignition delay is long, it will lead to fuel accumulation in the engine even before the ignition. When ignited, an explosion results as the combined effect of increased temperature and pressure. This is responsible for diesel knock. ✓

The diesel fuel should have a spontaneous ignition temperature less than the temperature produced by compression.

As the temperature to which air can be heated by compression is limited by various constraints, it is desirable to have fuels with short ignition delay, but the ignition delay must be long enough for the compression stroke to be completed. *Imp*

In order to grade the diesel fuels, cetane rating is employed.

Psi = pounds per square inch.

1 psi = 6894.757 Pascals
1 atm = 14.7 psi

Cetane Number

It is a

It is th

the same ign

Cetane $\text{C}_{16}\text{H}_{34}$
[or n-hexadecane] α -methyl

Cetane

For

As

ignite re

that high

alkyl nit

less knock

with a
and pr

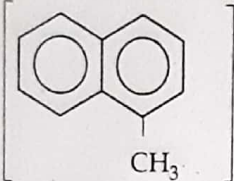
blend

Cetane Number

[GGSIPU, IInd Term, Nov. 2006]

It is a measure of the ease with which a fuel will ignite under compression. ✓

It is the % of hexadecane in a mixture of hexadecane and 1-methyl naphthalene which has the same ignition characteristics as the diesel fuel in question.

Diesel fuel	Cetane no.	Remarks
Cetane $C_{16}H_{34}$ [or n-hexadecane $CH_3(CH_2)_{14}CH_3$]	100	Very short ignition delay <i>so eliminate diesel knock.</i>
α -methyl naphthalene 	0	Longer ignition delay

Cetane no. of fuel primarily depends on the nature and composition of its hydrocarbons.

For instance, consider the following series :

(Highest cetane number) n -alkanes > naphthenes (i.e., cycloalkanes) > alkenes > branched alkanes > aromatics (lowest cetane number)

⇒ ignition delay increases from left to right

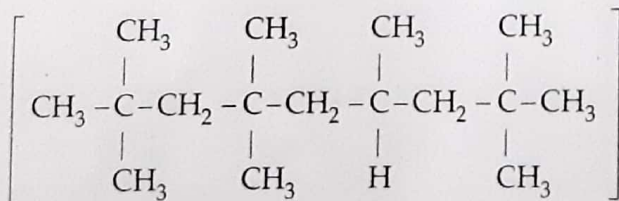
⇒ ignition quality increases from right to left

⇒ cetane no. increases from right to left.

As, straight chain alkanes like n -cetane have low ignition delay (high ignition quality) and ignite readily on compression, while aromatics do not ignite readily on compression, we can say that high cetane-no. fuels eliminate diesel knock. *gmp*

The cetane no. of diesel fuel may be raised by the addition of pre-ignition dopes such as alkyl nitrites such as ethyl nitrite, amyl nitrite etc.

2, 2, 4, 4, 6, 8, 8-hepta methyl nonane (HMN)



with a cetane rating of 15 is now considered as the low quality diesel in view of its easy availability and purity.

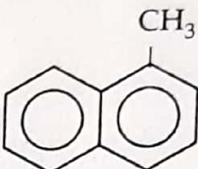
On the revised scale (HMN reference) the cetane number (CN) represents the % cetane in the blend with HMN plus $\frac{15}{100}$ of the % HMN.

Thus, a blend of 50% cetane and 50% HMN has a cetane rating of

$$50 + \frac{15}{100} \times 50 = 57.5.$$

The differences between the Octane number and Cetane number are summarized below :

[GGSIPU, IInd Term 2003, 05, 10 ; End Term 2001, 11]

S.No.	Octane No.	Cetane No.
1.	It is the % of iso-octane in a mixture of isooctane (2, 2, 4-trimethyl pentane) and <i>n</i>-heptane, which matches the fuel under test in knocking characteristics. e.g., an 60-octane fuel is one which has the same combustion characteristics as a 60 : 40 mixture of iso octane and <i>n</i>-heptane. $\text{CH}_3 - (\text{CH}_2)_5 - \text{CH}_3 ;$ <i>n</i>-Heptane Knocks very-badly (Octane no. = 0) $\begin{array}{c} \text{CH}_3 \quad \quad \text{CH}_3 \\ \quad \quad \\ \text{CH}_3 - \text{C} - \text{CH}_2 - \text{C} - \text{CH}_3 \\ \quad \quad \\ \text{CH}_3 \quad \quad \text{CH}_3 \end{array}$ <i>iso</i>-octane or 2,2,4-Trimethyl Pentane Gives very little knowcking (Octane no. = 100)	It is the % of hexadecane in a mixture of hexadecane and 1-methyl naphthalene which has the same ignition characteristics as the diesel fuel in question.  1-methyl naphthalen (Cetane no. = 0) $\text{CH}_3 - (\text{CH}_2)_{14} - \text{CH}_3$ <i>n</i>-Hexadecane (Cetane no. = 100)
2.	It is used for finding suitability of gasoline or any other internal combustion engine fuel.	It is used for finding suitability of a diesel fuel.
3.	The Octane number of I.C. fuels can be raised by the addition of tetraethyl lead (C ₂ H ₅) ₄ Pb or TEL and Diethyl telluride (C ₂ H ₅) ₂ Te.	The cetane number of diesel fuel can be raised by the addition of small quantity of certain 'Pre-ignition dopes' like ethyl nitrite, isoamyl nitrite, acetone peroxide etc.
4.	Hydrocarbons which are poor diesel fuels are quite good gasoline fuels.	Ignition quality order among hydrocarbon constituent of a diesel fuel is as follows : <i>n</i> -alkanes > naphthalenes > alkenes > branched alkanes > aromatics. Thus, hydrocarbons which are poor gasoline fuels are quite good diesel fuels.
5.	Straight chain hydrocarbon molecules are worst fuels. As they have low-Octane numbers.	Straight chain hydrocarbon molecules are best, as they have high cetane numbers.